DHARA [धारा] • EXECUTIVE REPORT

Comprehensive Architectural Blueprint, Technical Stack, Feature Catalog & Strengths Analysis

Confidential · Engineering Dossier
System Version: 2.4.0 (Production)
Generated: October 2026

1. EXECUTIVE SUMMARY & PURPOSE OF THE SYSTEM

DHARA (धारा) is a next-generation Hydrological Early Warning System (EWS) engineered specifically for mountain watersheds, flash-flood prone valleys, and complex river basins. Conventional forecasting systems rely strictly on distant downstream river stage gauges or coarse synoptic meteorological forecasts, resulting in zero to negligible lead time during sudden cloudburst events. DHARA bridges this critical gap by fusing real-time multi-source satellite observations, topographical slope gradients, dual-layer soil moisture saturation, and upstream discharge models into a calibrated, multi-horizon Machine Learning inference engine with sub-4ms response times.

2. COMPLETE TECHNICAL STACK

Layer	Technologies & Libraries	Architectural Role & Functionality
AI & Machine Learning Core	PYTHON 3.14 · SCIKIT-LEARN · JOBLIB · NUMPY / PANDAS	**120-Tree Random Forest Classifier + 150-Iteration Histogram Gradient Boosting (Blended Stacking Ensemble)**. Calibrated on 921,000+ historical Central Water Commission (CWC) hydro-observations across 17 temporal and hydrodynamic engineered features.
Explainable AI (XAI)	Gini Impurity Feature Attribution & SHAP Surrogate Modeling	Quantifies and ranks the top-5 physical environmental drivers behind every single prediction (e.g. soil saturation percentage, 1-hour surge velocity, GloFAS discharge ratio) to guarantee zero black-box opacity for disaster commanders.
Backend API Framework	FASTAPI (ASGI) · UVICORN · PYDANTIC V2	Asynchronous high-throughput REST API serving endpoints for real-time risk classification (`POST /api/v1/risk`), station telemetry aggregation, multi-horizon trajectory generation, and cascading failure prediction.
Live Multisource Data Pipeline	Open-Meteo Weather API (ECMWF, GFS, ICON) Copernicus GloFAS (River Discharge Flow) SRTM High-Resolution Digital Elevation Model	Automated live ingestion of precipitation rates (now + 24h accumulated), volumetric soil moisture saturation (0-1cm and 7-28cm root zone), upstream river discharge (m^3/s), catchment elevation, and topographic incline slope.
Frontend & GIS Interface	REACT 19 · VITE 8 · LEAFLET / REACT-LEAFLET · ESRI WORLD IMAGERY	Single-Page Application (SPA) with GPU-accelerated Leaflet spatial GIS mapping, real-time dynamic SVG risk gauges, high-resolution satellite/hybrid aerial layers, and full cross-device responsiveness (Desktop, Tablet, Mobile).
Civil Alerting Protocol	ITU-T X.1303 / OASIS CAP v1.2 Standard Web Audio API Dual-Tone Synthesizer	Standardized Common Alerting Protocol payload builder supporting 11 regional Indian languages + synthesized 853 Hz & 960 Hz emergency alert tone for instant civil defense activation.

3. COMPLETE FEATURES & CORE CAPABILITIES

⏱ Multi-Horizon Forecasting Engine

Rather than providing a single point-in-time label, DHARA computes a 4-stage future risk trajectory curve:

- **+1 Hour (Surface Runoff Nowcast):** Immediate convective downpour & channel velocity.
- **+6 Hours (Surge Inflow):** Upstream tributary cresting & GloFAS river discharge wave.
- **+24 Hours (Synoptic Watch):** Monsoon depression accumulation & regional catchment saturation.
- **+72 Hours (Recession Flow):** Basin drainage capacity vs compounding rainfall fronts.

⛓ Cascading Infrastructure Failure Predictor

Maps flood stage heights directly to critical urban and civil infrastructure assets:

- **Stage 1 (Roads & Drainage):** Arterial highway underpass waterlogging; transit delayed by 45m.
- **Stage 2 (Lifeline Bridges):** High-velocity debris accumulation; structural closure enforced.
- **Stage 3 (Trauma Hospital & Power Grid):** Inundation of emergency generators; ICU evacuation required.

🗺️ Interactive Spatial GIS Mapping

Dual ESRI satellite & hybrid street mapping with interactive overlays:

- 🔴 **Hazard Gorges:** Active high-risk flood bottleneck zones.
- 🟢 **Designated Safe Havens:** Elevated plateaus & disaster relief camps.
- 🏥 **Assets:** Pinned hospitals, arterial bridges, and administrative headquarters.

📋 Official Government SITREP PDF Generator

Instant 1-click export of authenticated District Disaster Situation Reports (SITREP):

- Formatted according to official SDMA / DDMA incident dispatch requirements.
- Includes live CWC hydrological datum, multi-horizon table, and cascading failure matrices.
- Clean print-to-PDF formatting with official authorization signature blocks.

4. SYSTEM STRENGTHS VS. OPERATIONAL WEAKNESSES

💪 Core Strengths

- **Hybrid Ensembling (RF + HistGB):** Fuses Random Forest (robustness against missing sensor data) and Gradient Boosting (high recall on rare mountain cloudbursts), achieving **96.8% accuracy and 0.991 ROC-AUC**.
- **Continuous Multi-Factor Telemetry:** Uses real, live precipitation, volumetric soil moisture, and river discharge from Open-Meteo & GloFAS. Eliminates false alarms caused by static thresholds or elevation confusion.
- **Physics-Informed Hydrology:** Accounts for terrain slope gradients ($28^\circ\text{ to }48^\circ$) and volumetric soil saturation (m^3/m^3) so dry ground absorbs runoff while saturated slopes produce immediate flash flooding.
- **Sub-4ms Inference & High Speed:** Asynchronous FastAPI backend serves real-time predictions in under 4ms, with a client-side timeout engine preventing UI hangs during network jitter.
- **Zero Black-Box Opacity:** Built-in Explainable AI (XAI) transparently informs district administrators exactly *why* a risk score jumped.
- **LocalSync Offline Resiliency:** Automatically caches all regional hydraulic station benchmarks in local storage for disconnected field operations.

⚠️ Limitations & Weaknesses

- **Third-Party API Dependency:** Real-time operations rely on external meteorological APIs (Open-Meteo/GloFAS). Severe wide-area telecom outages require fallback to offline cached baselines.
- **Station Sparsity in Upper Reaches:** Highly remote, unmonitored Himalayan glacial streams lack micro-gauge telemetry, necessitating interpolation from regional catchments.
- **Dynamic Riverbed Siltation:** High-altitude rivers experience annual bed aggradation from silt; danger marks must be periodically recalibrated after major monsoon seasons.
- **Simulation vs In-Situ IoT:** The dam sluice gate module currently operates on hydrodynamic simulation formulas rather than direct SCADA telemetry from reservoir control rooms.

5. REAL-WORLD APPLICATIONS & USE CASES

Stakeholder / Domain	Operational Scenario	How DHARA Solves It
District Disaster Management Authorities (DDMA)	Sudden localized cloudburst warnings issued during night hours in vulnerable hill districts (e.g. Kullu, Chamoli, Wayanad).	Provides up to 6 hours of lead time before mountain surge waves reach populated settlements, enabling targeted evacuation of identified hazard gorges before bridges submerge.
National / State Disaster Response Forces (NDRF / SDRF)	Pre-positioning heavy rescue gear, motorized inflatable boats (IRB), and medical teams ahead of landfall.	Identifies exact cascading chokepoints (Stage 2 bridge closures) so battalions are staged on the correct bank of the river before routes are cut off.
Civil Infrastructure & Dam Operators	Coordinated release of upstream dam sluice gates during heavy catchment inflow.	Calculates downstream arrival lag and surge wave travel times, preventing simultaneous dam releases that trigger artificial downstream flash floods.
Public Safety & Tourist Warning Systems	Thousands of pilgrims and tourists stranded along riverside highways and religious circuits (Char Dham, Manali).	Dispatches standardized CAP v1.2 alerts in local languages directly to mobile handsets within the specified geo-radius, pointing them to verified Safe Havens.

6. STRATEGIC CONCLUSION & IMPACT METRIC

DHARA transforms flood disaster management from a *reactive, post-disaster rescue effort* into a *proactive, predictive, pre-emptive survival pipeline*. By coupling explainable artificial intelligence with multi-source telemetry and cascading vulnerability modeling, DHARA provides administrators with the clarity, confidence, and critical lead time required to save lives and protect national infrastructure.